

Welcome to RASCAL!

Android User Manual

Version 1.8



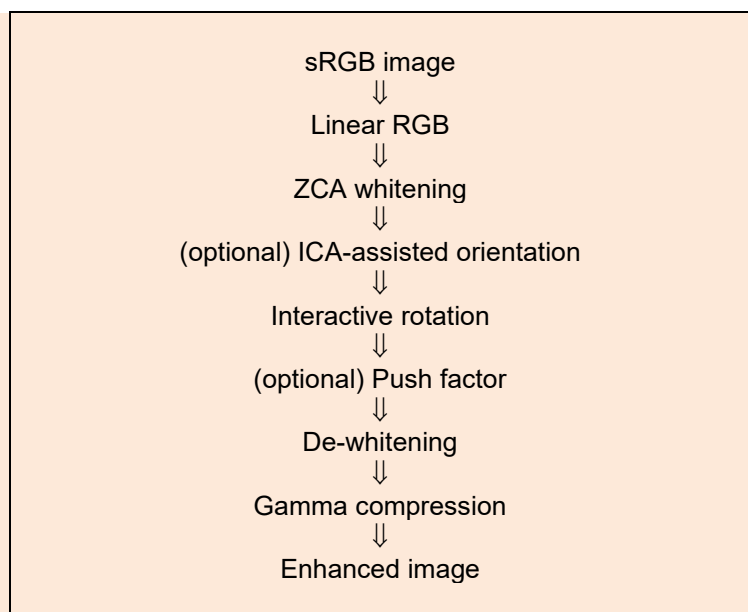
Edition	Android mobile application
Platform	Android 8.0+ (minSdk 26), compiled with SDK 35

1. Presentation

RASCAL (**R**adial **A**ngular **S**tatistical **C**olor **A**daptation and **L**ifting) is an interactive software designed to reveal subtle chromatic variations in standard RGB digital images. The program was developed primarily for applications in cultural heritage and archaeological imaging, where faint color differences may correspond to pigments, surface alterations, or archaeological structures that are difficult to detect visually. The application performs a decorrelation of RGB color channels followed by an interactive exploration of color transformations. This approach allows the user to visually enhance color contrasts and to concentrate relevant chromatic information on one or several of the image channels. Unlike fully automated enhancement techniques, RASCAL relies on seamless interactive exploration. The operator visually evaluates the transformations and identifies the most informative configuration.

2. Conceptual overview

Digital images are typically encoded in the sRGB color space, where the three-color channels may be strongly correlated. Such correlations often confine color information to a restricted region of the RGB cube, which reduces the visibility of some chromatic variations. RASCAL addresses this limitation using the following workflow:



After whitening in the linear RGB color space, the covariance matrix of the color distribution becomes spherical, meaning that all chromatic directions have unit variance. Because such a distribution is invariant under orthogonal rotations, the user can rotate the color space to align chromatic variations of interest with the RGB axes. Note that the whitening transformation is computed once from the original image (or from a subset of it) and remains fixed during the entire exploration. ICA-assisted orientation can also be chosen as starting point, as it may help separate statistically independent chromatic components. In any case, subsequent rotations do not modify the statistical model. As a result, the rotation performed by RASCAL does not correspond to a simple rotation of the RGB axes. Instead, the rotation operates in the whitened color space, whose geometry depends on the covariance structure of the image. Consequently, the transformation is image-dependent and reflects the internal geometry of the color distribution. This process often reveals

structures that are difficult or impossible to perceive in the original image. After rotation, the transformed data are mapped back to the original RGB space through the inverse whitening (de-whitening) transform, restoring the original variance structure. The resulting linear RGB values are then converted back to sRGB through gamma compression for display.

The detailed mathematical description of the method is provided in the 'Materials and Methods' section of the associated article.

3. Technical overview

The Android application is written in Kotlin and built as a native Android app. The project targets Android SDK 35, requires Android 8.0 or later (minSdk 26), enables ViewBinding, and uses CameraX for image capture. The rendering pipeline uses an OpenGL ES 2.0 GLSurfaceView so that rotations, zooming, and display updates remain interactive on the phone or tablet. Core dependencies include AndroidX core, AppCompat, Material Components, ConstraintLayout, ExifInterface, and CameraX. The application runs in full-screen mode and keeps the screen awake during use. Saved output images are written as JPEG files to Pictures/Rascal.

4. Installation

Since RASCAL is distributed as an APK file rather than through the Google Play Store, installing it requires a few manual steps. By default, Android blocks the installation of applications from unknown sources as a security measure. To install the APK, transfer the file to your device, then open it. Android will prompt you to allow installation from unknown sources. The exact wording varies by Android version: on Android 8.0 and later, you will be asked to grant the "Install unknown apps" permission to the app you are using to open the file (typically your file manager or browser). Tap **Allow** to proceed. Once the installation is complete, this permission can be revoked if desired. On some devices, a warning may also appear stating that the application is not verified by Google Play Protect. Tap **Install anyway** to continue.

Depending on the device manufacturer, additional security checks may be required. Some brands, such as Xiaomi or Huawei, add their own verification layer on top of standard Android, which may display extra warnings or require additional confirmation steps before the installation is allowed to complete. Simply follow the on-screen instructions and confirm each prompt.

5. Starting the application

When the application starts, an initial splash screen is shown. In this initial state, the main processing view is hidden and only the loading entry point is emphasized. Once an image has been loaded, the processing interface appears and the OpenGL preview becomes visible. The supported image sources are:

Take photo opens the dedicated camera workflow. Camera permission is requested when needed. After capture, the exposure and contrast can be adjusted before further processing. For GPS tagging, the app requests ACCESS_FINE_LOCATION and ACCESS_COARSE_LOCATION at runtime. When a photo is taken with the camera, the app automatically embeds the device's current GPS coordinates into the image's EXIF metadata.

A notification confirms whether the coordinates were successfully saved. If location permission has not been granted or no location is available, the photo is saved without GPS data.

Choose from gallery opens the system image picker. For gallery access, the app requests `READ_MEDIA_IMAGES` on recent Android versions and `READ_EXTERNAL_STORAGE` on older versions. The picker accepts image files in various formats: JPEG, PNG, BMP, and, on compatible devices, HEIF. The code explicitly warns that TIFF/RAW files are not supported and advises the user to use JPEG or PNG instead. EXIF orientation is corrected automatically after loading so that photographs appear with the proper rotation.

6. Loading an image

When an image is selected, the application first reads its dimensions, then decides whether downsampling is necessary. To reduce memory consumption and avoid crashes, very large images are downsampled so that the longest side does not exceed 4096 pixels. If downsampling occurs, the app informs the user. When a new image is manually loaded, the app resets the transformation parameters and displays the processing controls.

7. Interface description

The Android interface is simpler than the desktop version and is optimized for touch interaction. Its main elements are:

- A: Main image area: displays the processed image in real time.
- B: Top overlay: contains the loading action and related controls.
- C: Bottom overlay: contains transformation controls once an image is loaded.
- D: Rotation sliders: separate seek bars for X, Y and Z rotations.
- E: Preset buttons.
- F: Channel buttons: Color, R, G and B.
- G: ROI button: activates or deactivates ROI painting and ROI-based reprocessing.
- H: Reset, Split and Save buttons.
- I: Variance restoration and Push factor sliders.
- J: Random button.



8. Rotation and navigation

The user can explore the transformed color space either through sliders or through touch gestures.

Slider-based control

Three seek bars control rotation around the X, Y and Z axes. Internally, each slider is centered on a neutral position and converted to an angle by subtracting 180 degrees. Moving a slider immediately updates the renderer and redraws the image. This is the more precise procedure for fine adjustments.

Touch gestures

Gesture	Effect
1-finger drag	Rotates around X and Y. Horizontal motion changes Y rotation; vertical motion changes X rotation.
2-finger twist	Rotates around Z.
Pinch gesture	Zooms in or out around the focal point.
Double tap	Resets parameters; in ROI mode, launches ROI-based ZCA recomputation.

Zoom is applied relative to the pinch focal point, which makes navigation more comfortable on mobile devices. The **Reset** button also restores neutral rotation, sets the push factor back to its default value, resets zoom, clears any ROI overlay, and restores the original bitmap and original ZCA state.

The **Random** button launches an automatic sequence of random color-space orientations. A new random orientation is generated every 3 seconds, and the X, Y and Z sliders are updated accordingly. The sequence continues until the user interacts with any control, including, a touch gesture, or a pinch gesture, at which point it stops immediately. Tapping the **Random** button again starts a new sequence.

9. Split view

Tapping the **Split** button divides a 2D image panel into two comparison areas. The left side displays the original reference image, while the right side shows the current processed rendering, including the active ZCA rotation, variance restoration, or channel display mode. A white vertical divider marks the boundary between the two views. The divider can be dragged horizontally with the finger to adjust the relative size of the original and processed areas. When the image is zoomed or panned, the split remains aligned with the visible image area, making local before/after comparison easier.

10. Independent component analysis (ICA).

ICA-assisted orientation is an optional automatic orientation of the whitened color space. It is computed after ZCA whitening by tapping the **ICA** button, using a sample of up to 50,000 pixels. Once the ICA orientation has been applied, the ICA button turns yellow. ICA does not replace manual exploration; it simply proposes an orientation of the whitened color space that may help separate statistically independent chromatic components. The user can then continue the exploration manually using the rotation sliders or touch gestures. As soon as the rotation is manually modified, the **ICA** button returns to white, indicating that the current orientation is no longer the one computed by ICA.

11. Channel visualization

Four display modes are available:

Color: full processed RGB image.

R: red-channel grayscale view.

G: green-channel grayscale view.

B: blue-channel grayscale view.

When a mode is selected, the corresponding button is highlighted in red.

12. ROI (Region Of Interest) processing

The Android app includes a touch-based ROI workflow. Pressing the **ROI** button enters painting mode. In this state, the user paints directly on top of the image with a semi-transparent yellow brush the area expected to contain chromatic elements of interest. This option allows color enhancement to focus on specific areas of interest in the original image while excluding external elements whose colors differ markedly from those of the region under investigation. The brush is approximately equivalent to a 60-pixel stroke, corresponding to a radius of about 30 pixels.

The ROI workflow is as follows:

- Press ROI to enter paint mode.
- Paint the region of interest with one finger.
- Double tap to process the painted region.
- Press ROI again to exit paint mode or to cancel an active ROI result.

When ROI processing is launched, the app recomputes the ZCA parameters from the selected pixels from the original image. If the ROI is too small, processing is refused and the user is prompted to paint a larger area. If ICA is activated after a ROI, ICA is only computed from that selected region rather than from the full image. In that case, up to 50,000 pixels are sampled from the ROI, and the resulting orientation is applied to the rotation sliders, after which it can still be adjusted manually, as previously.

13. Presets

Presets allow you to save and instantly recall transformation states while working on an image. When a new image is loaded, Preset 1 is automatically initialized with the original transformation state, making it easy to switch back and forth between the original view and processed states. To save the current state as a preset, tap the **Preset** button. The app stores the current rotation values, push factor, channel mode, Var and, if a ROI has been applied, the full ROI transformation. Up to five presets can be saved at a time. Each saved preset appears as a numbered button (1–5). To recall a preset, tap its number. The sliders and channel mode are immediately restored. If the preset includes a ROI transformation, the image is also updated accordingly. Tap the **X** button to clear all stored presets at once, except the first one corresponding to the original image. Presets are also automatically cleared when a new image is loaded. Note that presets are session-only: they are not saved between app restarts.

14. Variance Restore

The **Var** slider controls how much of the original color variance is restored after whitening.

At 1.00 (default), the full inverse whitening transform is applied. The original variance structure is restored, producing the standard RASCAL rendering with more visually stable colors.

At 0.00, variance restoration is disabled. The decorrelated and variance-equalized color components are mapped directly for display, producing a stronger DStretch-like effect, with enhanced chromatic contrast and less natural colors.

Intermediate values provide a continuous transition between standard RASCAL rendering and a stronger decorrelation-stretch-like visualization.

15. Push factor

The **Push** factor is controlled by a dedicated seek bar. The displayed label is updated continuously. The default state corresponds to 1.0. Values above 1.0 amplify the transformed effect, whereas values below 1.0 attenuate it.

16. Saving processed images

The Save button captures the processed result and writes it to the device gallery. Files are saved as JPEG images with quality 95 and a timestamp-based filename in the form RASCAL_YYYYMMDD_HHMMSS_processed.jpg. If saving fails, the application reports the error to the user through a toast message.

17. Limitations of the current Android version

Compared with the desktop manual provided with the project, the Android version documented here does not expose several desktop-only functions: 3D model processing, 3-D color cloud visualization, expert/basic

mode switching, radial push mode, export of multiple view panels, or session save/load. The present manual therefore focuses strictly on the Android features. DNG/RAW files are not supported, and very large images may be downsampled automatically.

18. Quick reference

Action	How to do it
Load an image	Tap Load , then choose Camera or Gallery.
Rotate X/Y	Use the X and Y sliders or drag with one finger.
Rotate Z	Use the Z slider or twist with two fingers.
Zoom	Pinch on the image.
Activate split view	Tap Split . Drag the vertical divider with one finger to adjust the comparison area.
Compute ICA rotation	Tap ICA .
Random rotation	Tap Random . Any manual interaction immediately stops the sequence.
Change push factor	Move the Push slider.
Adjust variance restoration	Move the Var slider.
Inspect one channel	Tap C , R , G , or B .
Store the current configuration	Tap Preset
Clear presets	Tap x .
Paint an ROI	Tap ROI , paint with one finger, then double tap to process.
Cancel ROI	Tap ROI again or reset the parameters.
Reset the view	Tap Reset or double tap outside ROI mode.
Save result	Tap Save ; the file is written to Pictures/Rascal.

19. Final note

RASCAL is an exploratory visualization tool. Processed images should always be interpreted in the scientific, archaeological, or documentary context of the original data.

20. Contact

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